

# Solvent-Based Lithium Extraction: A Combined Molecular Dynamics and Experimental Study

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## 01 INTRODUCTION

The rapid proliferation of lithium-ion batteries (LIBs) in energy storage, electric vehicles, and the renewable energy domain has highlighted the necessity of sustainable recycling techniques to mitigate supply chain and environmental risks. Recycling lithium from spent batteries enables the reintroduction of valuable resources into the supply chain for transitioning towards a circular economy and sustainable development [1] [2]. Hydrophobic deep eutectic solvents (HDES) can play a key role in enabling their direct usage in liquid-liquid extraction.

## 02 OBJECTIVES

- ✓ To formulate HDES for efficient extraction of lithium from aqueous solution.
- ✓ To understand the atomistic extraction, ion mobility mechanisms and interfacial behavior using molecular dynamics (MD) simulations.
- ✓ To validate the experimental results with the MD simulations.

## 03 METHODOLOGY

- ✓ Force fields: OPLS4
- ✓ DESMOND MD (Schrodinger Material Science Suite)
- ✓ Water model: Simple Point Charge (SPC)
- ✓ Interface: (LiCl in Water) + HDES
- ✓ NVT production run of 100 ns at 300 K

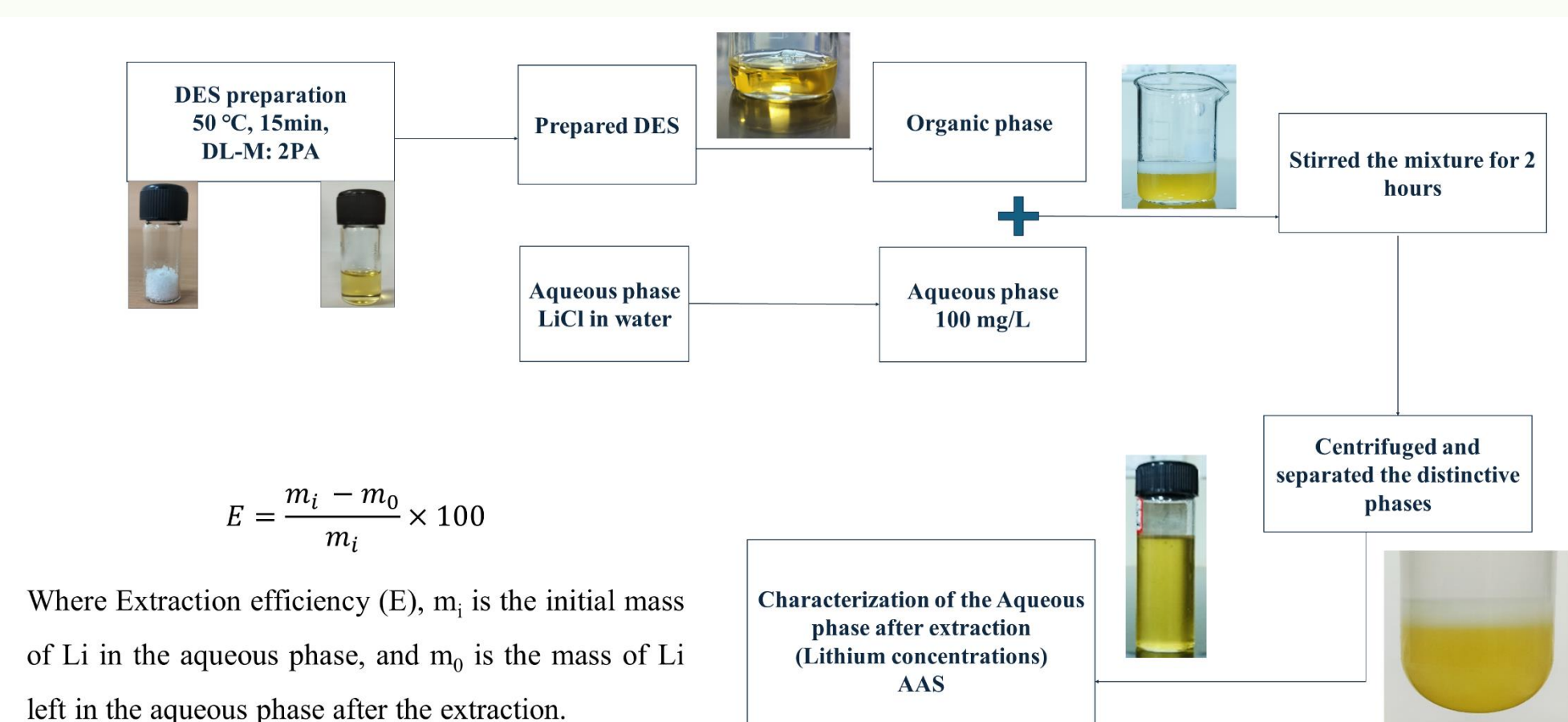


Figure 1. Liquid-liquid extraction of Li experimental protocol.

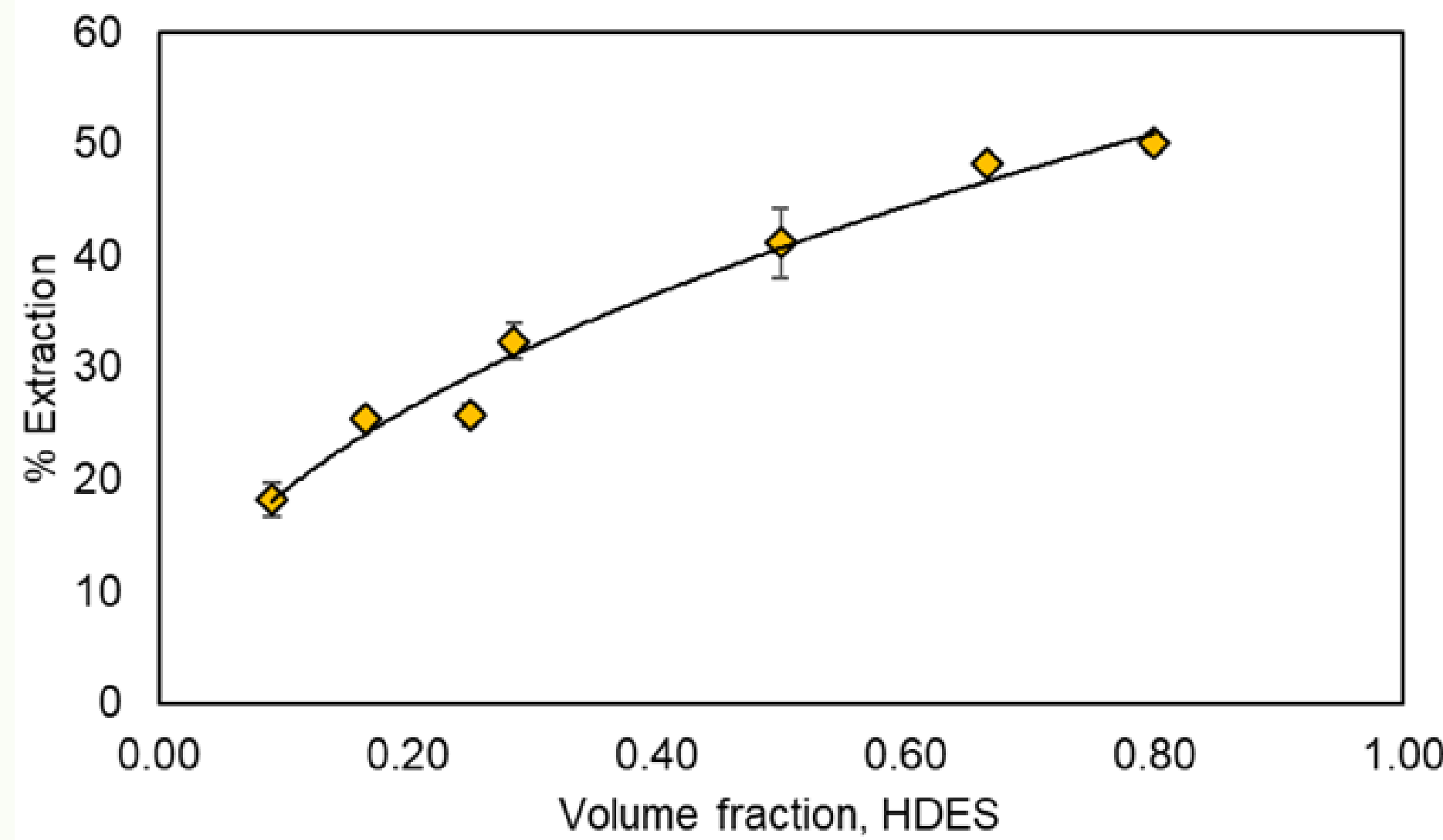


Figure 2. Effect of volume fraction of HDES on the extent of Li extraction.

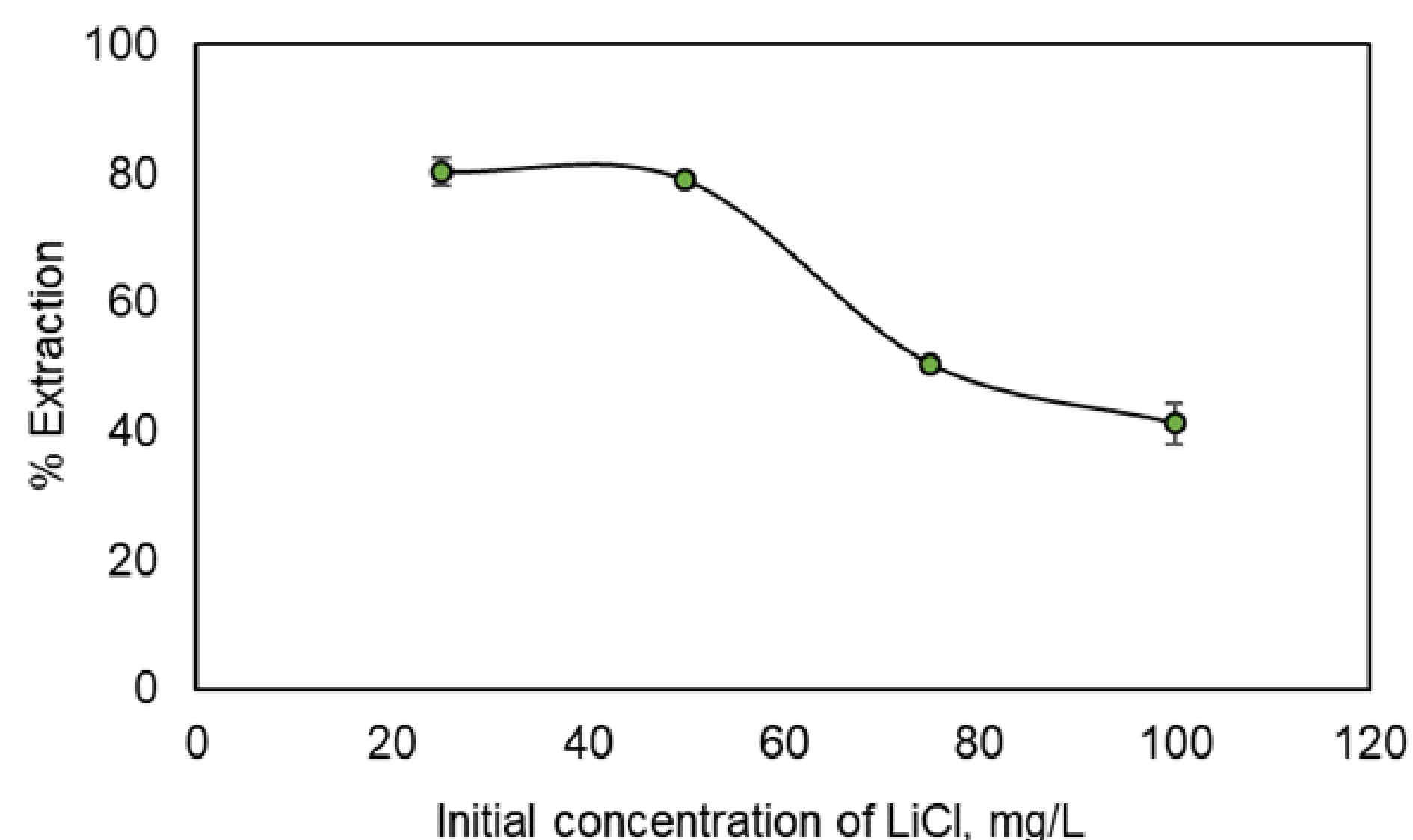


Figure 3. Effect of initial concentration of LiCl on the extent of Li extraction.

## 04 RESULTS

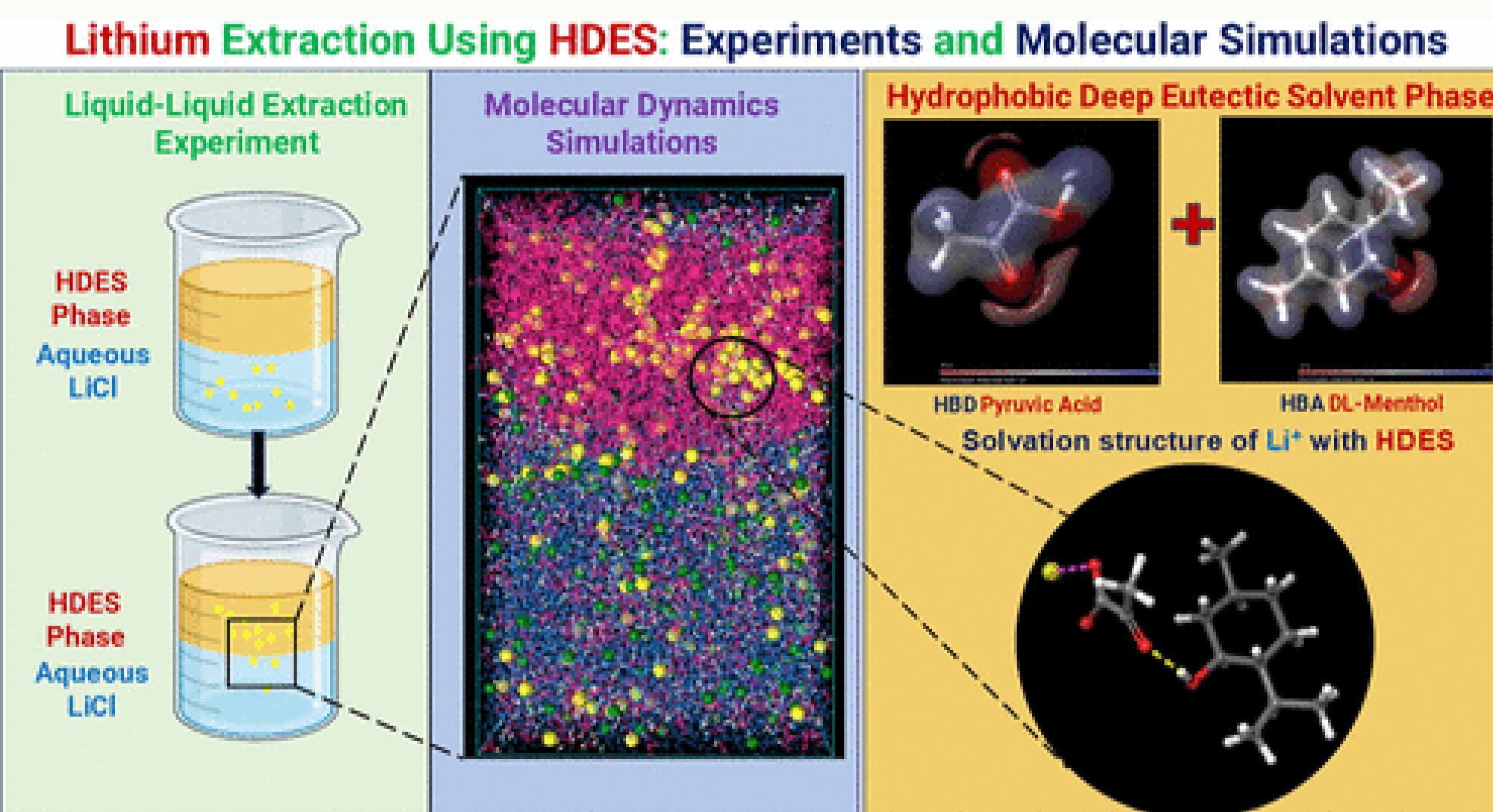


Figure 4. Combined molecular dynamics and experimental study: Graphical abstract

Table 1. Calculated  $pK_a$  values.

| HDES   | DL-Menthol | Pyruvic Acid |
|--------|------------|--------------|
| $pK_a$ | 14.42      | 2.32         |

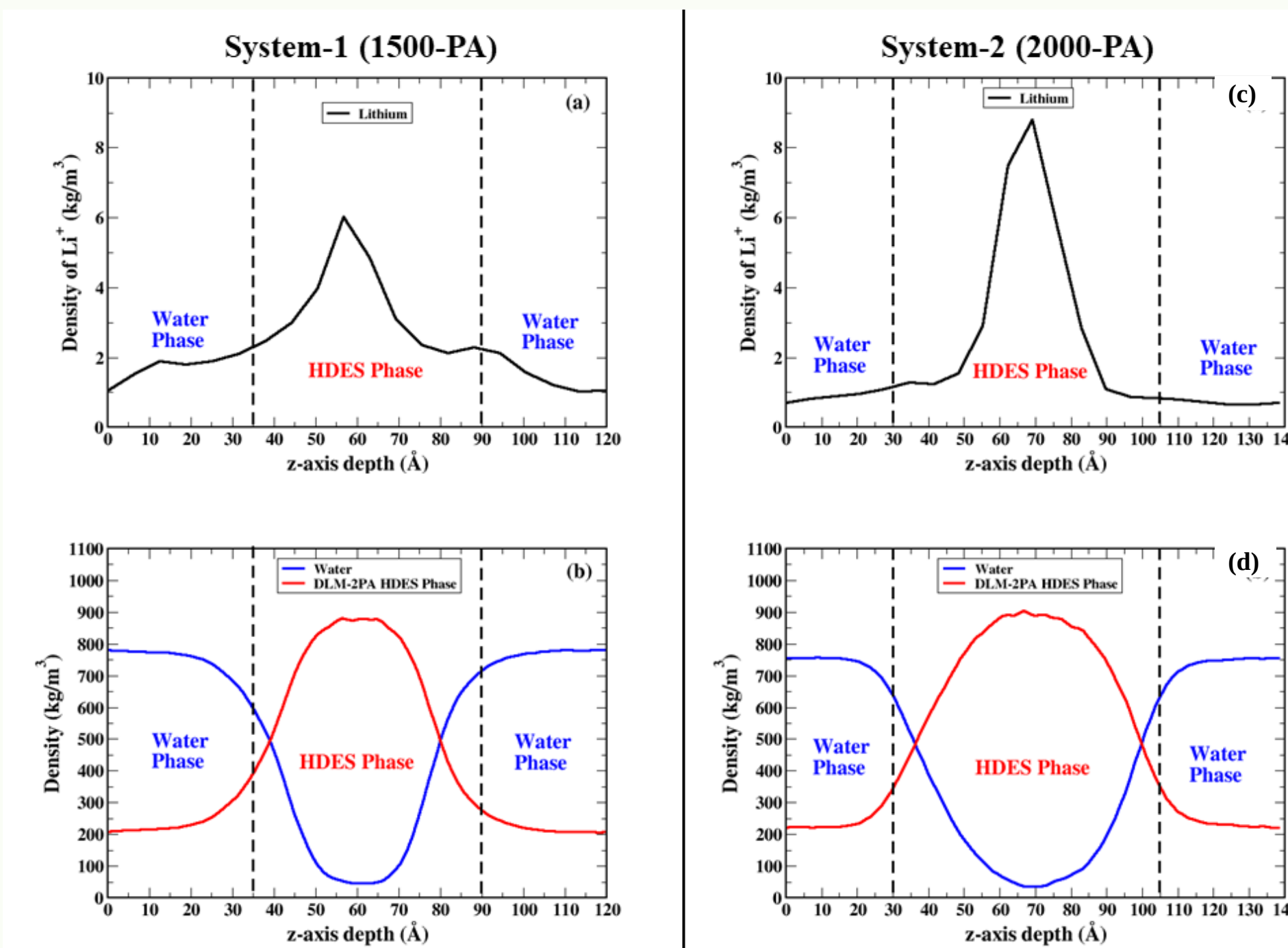


Figure 5. Density profile of a) and c)  $Li^+$  ions, b) and d) HDES and aqueous phase along the interface. Black line for lithium, red line for HDES, and blue line for water. Density in  $kg/m^3$ .

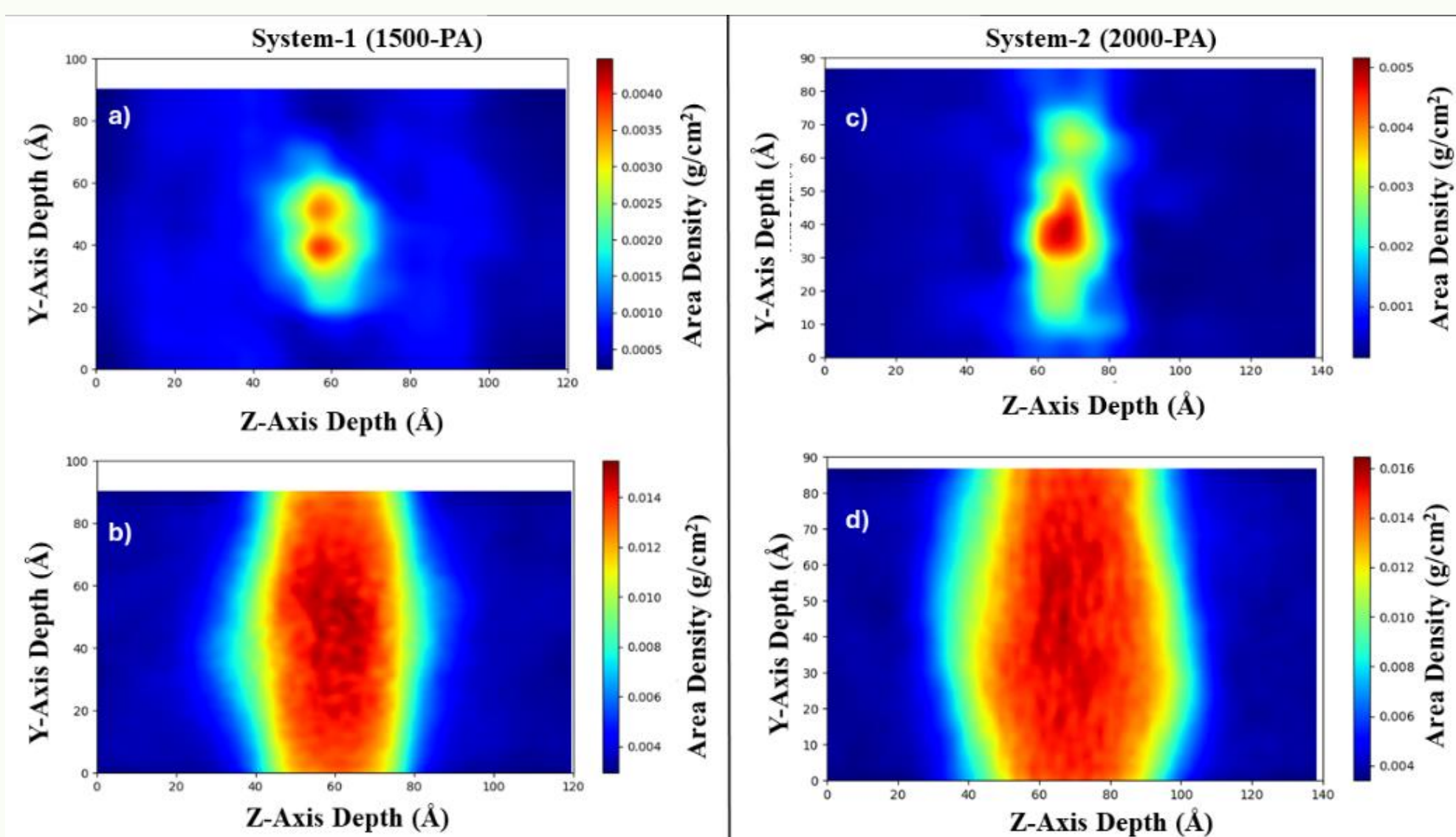


Figure 6. Density profile of a) and c)  $Li^+$  ions, b) and d) HDES and the aqueous phase along the interface. The dark red color represents the high HDES density (in  $g/cm^2$ ) in (b, d), while the dark blue color represents the low HDES density (in  $g/cm^2$ ) in (b, d).

Table 2. Calculated Diffusion Coefficient of  $Li^+$  ion.

| Interphase         | Diffusion Coefficient of $Li^+$ ( $m^2/s$ ) | Standard deviation ( $m^2/s$ ) |
|--------------------|---------------------------------------------|--------------------------------|
| System-1 (1500 PA) | $1.16 \times 10^{-10}$                      | $3.95 \times 10^{-14}$         |
| System-2 (2000 PA) | $0.42 \times 10^{-10}$                      | $4.58 \times 10^{-14}$         |

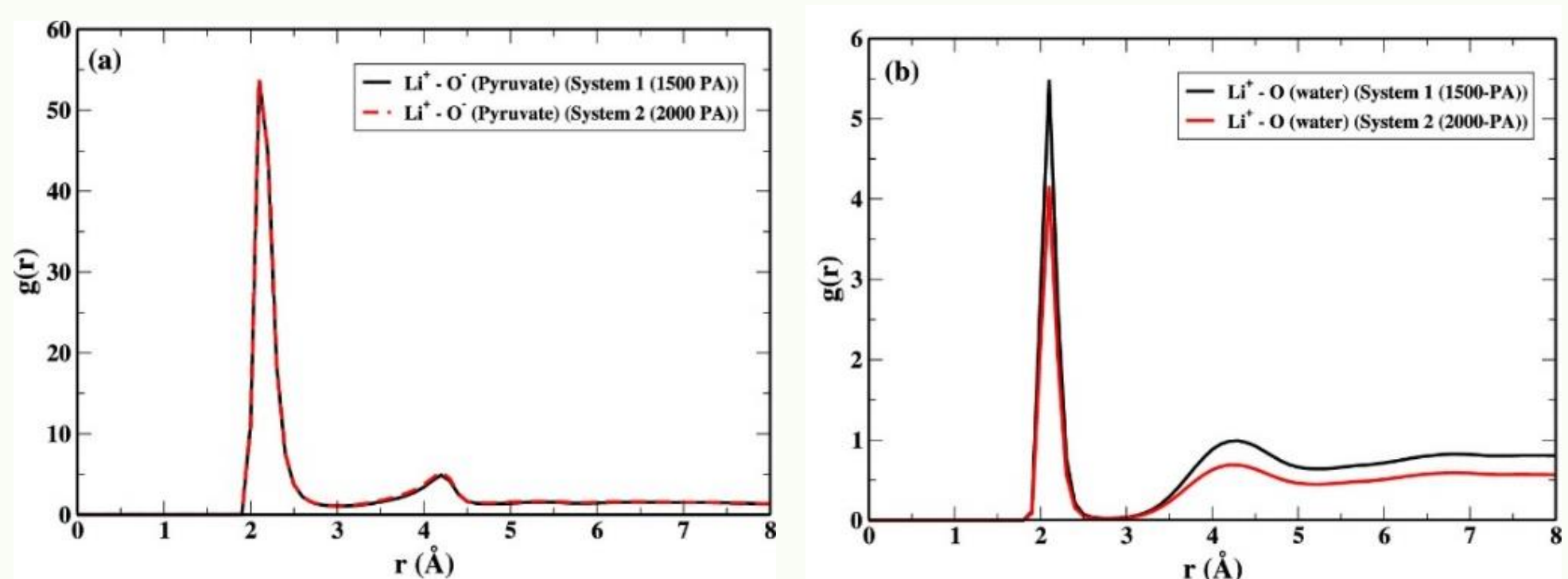


Figure 7. RDF plot for a)  $Li^+ - O^-$  (HDES), b)  $Li^+ - O$  (water).

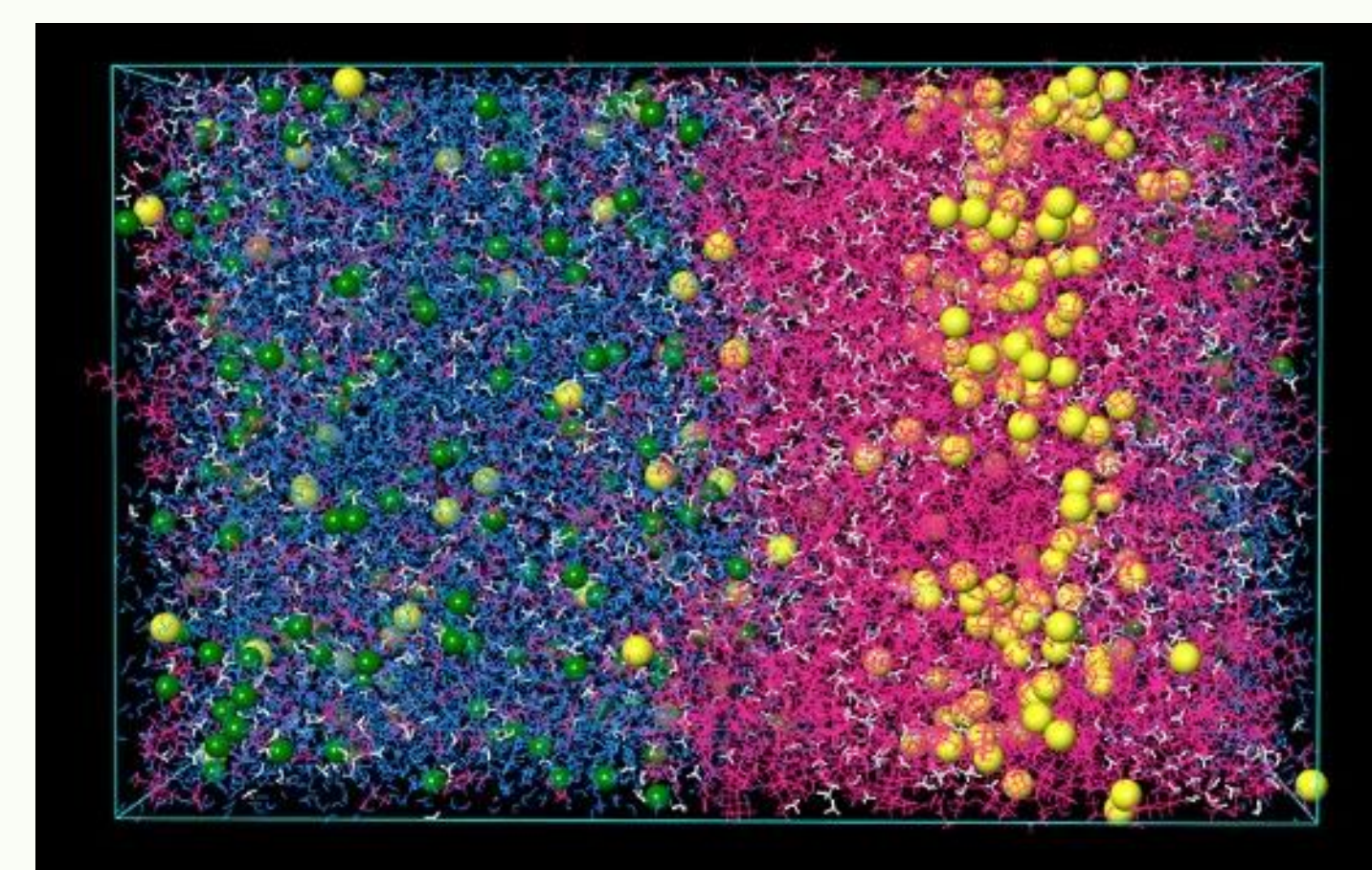


Figure 8. Snapshots of the LiCl in water-HDES interface of MD NVT run,  $Li^+$  (yellow),  $Cl^-$  (green), HDES (pink), water (blue),  $H_3O^+$  (white).

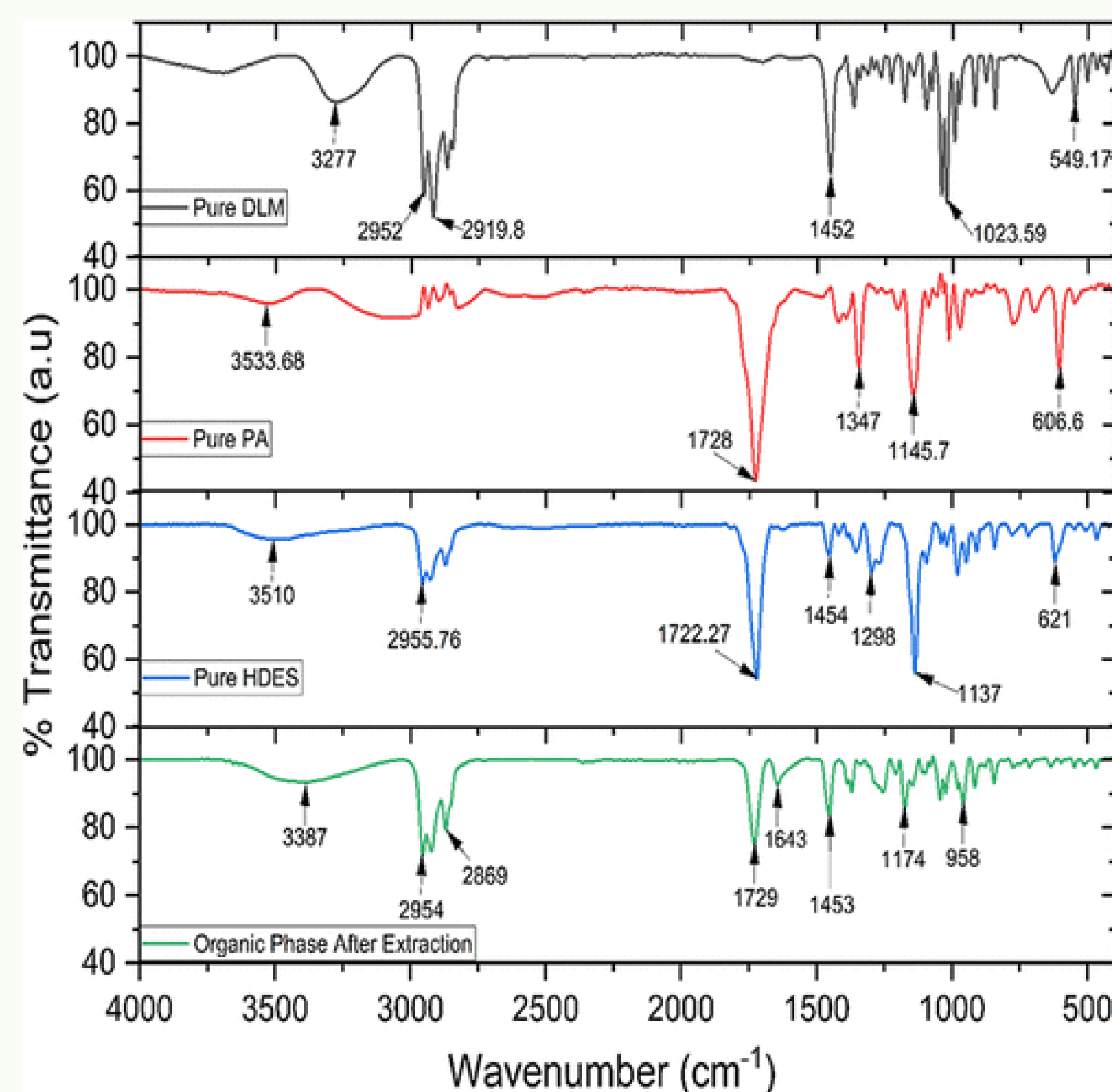


Figure 9. FTIR plot for pure DLM, PA, HDES and organic phase after lithium extraction.

## 05 CONCLUSIONS

- The formulated menthol-based HDES has affinity towards lithium.
- The highest extraction efficiency of 80 % was achieved.
- The lithium extraction efficiency significantly improved with the increase in the HDES volume fraction in the mixture.

The influence of HDES volume fraction on lithium extraction was validated by MD simulation.

- KEY FINDING

## 06 FUTURE WORK

Leaching of lithium, cobalt, nickel, and manganese from the spent cathode materials using Deep Eutectic Solvents.

## 07 REFERENCES

- Shamanth Y U, et al. Extraction of Lithium from an Aqueous Solution Using a Hydrophobic Deep Eutectic Solvent: A Combined Experimental and Classical Molecular Dynamics Study. Industrial & Engineering Chemistry Research. 2025 Dec 30, 65.1:799-808.
- Shamanth Y U, et al. Lithium extraction using ionic liquids: Insights from quantum chemical and molecular dynamics simulations. Journal of Ionic Liquids. 2025 Oct 15:100177.

## 08 ACKNOWLEDGEMENTS

- We would like to express gratitude for the support provided by the Schrodinger Centre for Molecular Simulations, Manipal Academy of Higher Education (MAHE), Manipal.
- We would like to express gratitude for the support provided by the MAHE Manipal for Intramural Funding (MAHE/CDS/PHD/IMF/2023) and Dr. TMA PAI Scholarship.